



# BACTERIAL DISEASES — FULL MIND MAP

Biology Ch. 9.ii · Diseases & Treatment · Hinglish · SSC / BPSC / BSSC / Railway



## Topics in this sub-chapter:

1. Food Poisoning — Salmonella & Botulism
2. Anthrax, Diphtheria & DPT Vaccine
3. Haemophilus Influenzae & Pneumococcal Vaccine
4. Tuberculosis — Cause, Test, BCG & Treatment
5. Tetanus, Plague & Transmission Routes
6. Dental Caries & Water-borne Diseases
7. Leprosy & Typhoid
8. Antibiotics — Waksman, Penicillin & Broad Spectrum
9. Sulpha Drugs & Antibacterial Agents
10. Multi-drug Resistance (MDR)

## 1. Food Poisoning — Salmonella & Botulism

- Food poisoning source mein Salmonella bacilli se linked hai; Salmonella intestinal bacterial infection group hai aur contaminated food/water se spread ho sakta hai. [Q1]
- Botulism Clostridium species—especially Clostridium botulinum—ke toxin se linked serious bacterial illness hai; improperly canned/preserved food aur wounds source routes mein diye gaye hain. [Q2]
- Botulism ko food-borne intoxication kaha gaya hai; toxin nerves ko affect karke breathing difficulty, muscle paralysis aur severe cases mein death tak cause kar sakta hai. [Q3]

## 2. Anthrax, Diphtheria & DPT Vaccine

- Splenic fever = Anthrax; source ise bacterial infection batata hai jo livestock ko affect kar sakta hai aur humans mein bhi transmit ho sakta hai. [Q4]
- Diphtheria vaccine toxoid type ki vaccine hai aur disease Corynebacterium diphtheriae se linked hai. [Q5]
- DPT ka full protective set = Diphtheria + Pertussis (Whooping Cough) + Tetanus; source ke multiple questions isi triple-antigen combination ko repeat karte hain. [Q6, Q7, Q8, Q10]
- DPT vaccine Polio ke liye use nahi hoti. [Q9]

## 3. Haemophilus Influenzae & Pneumococcal Vaccine



### EXAM TRAP — Haemophilus influenzae — Multiple Correct

- ▶ Q11 mein source correct answers A & D deta hai: Haemophilus influenzae young children mein meningitis aur pneumonia dono cause kar sakta hai. Influenza virus se hota hai aur whooping cough Bordetella pertussis se. [Q11]

- Pneumococcal Conjugate Vaccine Streptococcus pneumoniae diseases se protection ke liye hai; source meningitis, sepsis aur bacteremic pneumonia ke against effectiveness aur antibiotic-resistant disease burden reduction mention karta hai. Statements 1 & 2 correct, statement 3 incorrect because side effects/allergic reactions possible hain. [Q12]

## 4. Tuberculosis — Cause, Test, BCG & Treatment

- Measles, Rabies aur Polio viral diseases hain, jabki Tuberculosis bacterial disease hai aur source Mycobacterium tuberculosis ko cause batata hai. [Q13, Q18, Q19]
- Source ke mutabik P-amino salicylic acid antimicrobial drug tuberculosis aur leprosy dono ke treatment mein suitable hai. [Q14]
- WHO-context questions mein Tuberculosis ko present-time major cause of death aur India mein lakhs of annual deaths se linked disease bataya gaya hai. [Q15, Q16]

- Genetic disease—Tuberculosis incorrect match hai; source TB ko bacterial disease batata hai, genetic disease nahi. [Q17]
- TB detection ke liye source spelling 'Montoux Test' deta hai; explanation ise Tuberculin Skin Test/TST ke context mein describe karta hai. [Q20]
- BCG (Bacillus Calmette-Guerin) vaccine Tuberculosis prevention ke liye hai; source newborn ko just after birth inoculation best batata hai aur up to 12 months possible hone ka note deta hai. [Q21, Q22, Q23]
- Matching question mein Malaria—Chloroquine correct pair hai; source BCG→TB, ATS→Tetanus aur Scurvy→Vitamin C context bhi explain karta hai. [Q24]

## 5. Tetanus, Plague & Transmission Routes

- Tetanus Clostridium tetani bacterial disease hai aur 'Lock jaw' naam se bhi known hai; wound entry ke baad toxin painful muscle tightening aur jaw/neck locking cause kar sakta hai. [Q25]
- Plague infected mosquito bite se caused nahi; source Yersinia pestis ko cause aur infected fleas/rodents ko transmission chain batata hai. [Q26]
- Transmission match: Air→T.B.; Water→Cholera; Contact→Syphilis; Wound→Tetanus. [Q27]
- Disease-feature match: Plague→flea-bites; Filaria→mosquitoes; Beri-beri→Vitamin B deficiency; Typhoid→intestines. [Q28]

## 6. Dental Caries & Water-borne Diseases

- Tooth decay/caries mainly mouth bacteria + fermentable carbohydrate food particles ke interaction se linked hai; source caries formation mein tooth surface, bacteria, carbohydrates aur time ko key factors batata hai. [Q29]
- Dental caries ka reason source mein bacterial infection diya hai. [Q30]
- Shallow handpump-water question mein source Cholera, Typhoid aur Jaundice risk batata hai, jabki Fluorosis ko excess fluoride condition aur exception mark karta hai. [Q31]
- Cholera Vibrio cholerae bacterial intestinal disease hai aur contaminated water/food se spread hota hai; Typhoid aur Cholera typical water-borne diseases diye gaye hain. [Q32, Q33]

## 7. Leprosy & Typhoid

- Leprosy/Hansen's disease bacterial disease hai; source Mycobacterium leprae ya Mycobacterium lepromatosis ko causes batata hai. [Q34]
- Typhoid Salmonella typhi bacteria se caused hai; source question mein Jaundice, Influenza aur Mumps ko viral association ke contrast mein diya gaya hai. [Q35]

## 8. Antibiotics — Waksman, Penicillin & Broad Spectrum

- 'Antibiotic' term source ke mutabik Selman Waksman ne 1942 mein coin kiya; Waksman streptomycin aur several antibiotics discovery se bhi linked hai. [Q36]

### ⚠ EXAM TRAP — More Than One Antibiotic

- ▶ Q37 source answer (E) More than one: Sulphaguanidine aur Chloramphenicol dono antibacterial/antibiotic agents diye gaye hain. [Q37]
- ▶ Q38 source answer (E) More than one: Penicillin aur Sulphadiazine dono antibiotic agents diye gaye hain. [Q38]

- Penicillin ek well-known antibiotic medicine hai. [Q39]
- Chloramphenicol source mein broad-spectrum antibiotic hai—Gram-positive aur Gram-negative dono bacteria ke wide range par act karta hai aur serious infections including typhoid context mein mentioned hai. [Q40]

## 9. Sulpha Drugs & Antibacterial Agents

- Sulfa drugs bacteria-caused diseases ke against effective aur primarily antibacterial class ke drugs hain; source sulfonamide examples aur Prontosil ko first sulfa drug (1932) mention karta hai. [Q41, Q42]
- Cibazole source mein short-acting sulpha drug hai; Penicillin/Streptomycin antibiotics aur Aspirin antipyretic comparison ke roop mein diye gaye hain. [Q43]

## 10. Multi-drug Resistance (MDR)

- India mein microbial multi-drug resistance ke source-listed reasons mein incorrect antibiotic doses aur livestock farming mein antibiotics ka use include hain; given statements mein 2 & 3 correct hain. [Q44]